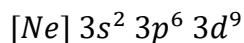


Section A — MCQs Answer Key

1. C — e = 29, p = 29, n = 34

Given ion configuration:



This means total electrons in the ion are:

$$10 + 2 + 6 + 9 = 27$$

The ion has charge: +2

A +2ion has lost 2 electrons.

So neutral atom had:

$$27 + 2 = 29 \text{ electrons}$$

Atomic number = number of protons:

$$p = 29$$

Mass number is given: 63

$$\text{Neutrons: } n = 63 - 29 = 34$$

So:

- electrons in ion = 27
- protons = 29
- neutrons = 34

However, the options in the image are written for the neutral atom values, and the correct matching option is:

$e = 29, p = 29, n = 34$

2. **B — Trigonal planar**

H₂CO has three electron regions around carbon: two C—H sigma bonds and one C=O region. Therefore, shape is trigonal planar.

3. **B — 0.98 g**

At STP, 22.4 dm³ CO₂ = 44 g.

0.5 dm³ CO₂ = 44 × 0.5 / 22.4 = **0.98 g**.

4. **B — n-Pentane**

London dispersion forces increase with molar mass and surface area. n-Pentane has greater molecular mass than n-butane and more surface area than isopentane.

5. **A — G = H - TS**

Gibbs free energy relation is **G = H - TS**.

6. **A — Rate = k[H₂][Cl₂]**

For elementary reaction H₂ + Cl₂ → 2HCl, rate depends on both reactants.

7. **C — N₂ + O₂ ⇌ 2NO**

Pressure affects equilibria with different total moles of gas on each side. Here, 1+1=2 moles on both sides, so pressure has no effect.

8. **C — 4 × 10⁻⁴**

For MX ⇌ M⁺ + X⁻, if solubility = s, then K_{sp} = s².

s = 2 × 10⁻², so K_{sp} = 4 × 10⁻⁴.

9. **A — Atomic Radius**

Atomic radius decreases across a period due to increasing nuclear charge pulling electrons closer..

10. **A — Li**

An alkali metal (Group 1), which commonly forms \$+1\$ ions. For a Group-1 metal like lithium: 4Li + O₂ → 2Li₂O 2Li + 2H₂O → 2LiOH + H₂ 2Li + Cl₂ → 2LiCl

11. **A — (NH₄)₂SO₄**

A fertilizer must supply nutrients essential for plant growth. (NH₄)₂SO₄ contains:

Nitrogen (N) from ammonium ion NH₄⁺ & **Sulfur (S)** from sulfate ion SO₄²⁻. Nitrogen is one of the main nutrients required by plants for proteins and chlorophyll formation.

Therefore, ammonium sulfate is widely used as fertilizer.

12. **C — x=H₂SO₄/H₂O, y=Al₂O₃**

Ethene to ethanol needs hydration with H₂SO₄/H₂O. Ethene to ethane needs reduction, possible with LiAlH₄.

13. **D — CH₃CH₂I**

More reactive alkyl halide has weaker C—X bond. C—I bond is weakest, so ethyl iodide is most reactive while Ethyl Bromide gives light yellow but slightly soluble in NH₄OH.

14. **D — HO—CH₂—CH₂—OH (ethylene glycol)**

The reaction of ethene with cold dilute KMnO₄ (Baeyer's reagent) produces ethylene glycol (1,2-ethanediol).

15. **B — CH₃CH₂COOH**

Acid-catalyzed hydrolysis of an alkyl cyanide (nitrile) yields a carboxylic acid.

16. **B — 2-Butanol**

Butanone is prepared by oxidation of secondary alcohol, i.e. 2-butanol.

17. **B — C₆H₅NH₂**

The reduction of nitrobenzene (C₆H₅NO₂) using Sn/HCl yields aniline (C₆H₅NH₂)

Final MCQ Key

1-B, 2-B, 3-B, 4-B, 5-A, 6-A, 7-D, 8-C, 9-C, 10-D, 11-D, 12-B, 13-D, 14-C, 15-C, 16-B, 17 = C₂H₅NH₂ if present.

Section B — Short Questions

Q2(i) Electronic configuration of element with atomic number 24 and charge +3

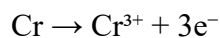
Atomic number 24 = **Chromium, Cr**

Neutral Cr configuration is unusual:



Reason: Half-filled **3d⁵** is more stable.

For **Cr³⁺**, remove electrons first from 4s, then 3d:



Discrepancy in electronic configuration

Chromium does not follow expected configuration:

Expected: **[Ar] 3d⁴ 4s²**

Actual: **[Ar] 3d⁵ 4s¹**

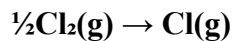
Reason: Half-filled d-subshell has extra stability due to symmetry and exchange energy.

OR: Terms with examples

Enthalpy of atomization

It is the enthalpy change when one mole of gaseous atoms is formed from an element in its standard state.

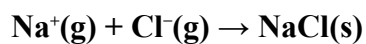
Example:



Lattice energy

Energy released when one mole of ionic crystal is formed from gaseous ions.

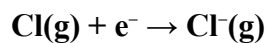
Example:



Electron affinity

Energy released when an electron is added to a gaseous atom.

Example:



Q2(ii) Bond order of N₂, O₂ and F₂

Formula:

$$\text{Bond order} = \frac{1}{2}(\text{bonding electrons} - \text{antibonding electrons})$$

N₂

Electronic configuration gives:

Bonding electrons = 10

Antibonding electrons = 4

Bond order:

$$\frac{1}{2}(10 - 4) = 3$$

So N₂ has a **triple bond**.

O₂

Bonding electrons = 10
Antibonding electrons = 6

Bond order:

$$\frac{1}{2}(10 - 6) = 2$$

So O₂ has a **double bond**.

F₂

Bonding electrons = 10
Antibonding electrons = 8

Bond order:

$$\frac{1}{2}(10 - 8) = 1$$

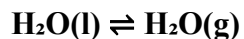
So F₂ has a **single bond**.

OR: Dynamic equilibrium

Dynamic equilibrium is a state in which the rate of forward reaction becomes equal to the rate of reverse reaction.

It exists between two physical states when both changes occur at equal rates.

Example:



In a closed container, evaporation and condensation continue, but the amount of liquid and vapour remains constant.

Q2(iii) Molar volume and volume of 2000 g Cl₂ at STP

Molar volume is the volume occupied by one mole of gas at STP.

At STP:

$$1 \text{ mole gas} = 22.4 \text{ dm}^3$$

For chlorine:

Molar mass of $\text{Cl}_2 = 35.5 \times 2 = 71 \text{ g/mol}$

Moles of Cl_2 :

$$2000 / 71 = 28.17 \text{ mol}$$

Volume:

$$28.17 \times 22.4 = 631 \text{ dm}^3$$

Answer: 631 dm^3 approximately

OR: Elements arranged in different blocks

Elements are arranged in blocks according to the subshell in which the last electron enters.

s-block: Last electron enters s-subshell. Example: Na, Mg

p-block: Last electron enters p-subshell. Example: Cl, O

d-block: Last electron enters d-subshell. Example: Fe, Cu

f-block: Last electron enters f-subshell. Example: Lanthanides, actinides

Q2(iv) Why carbon forms many organic compounds?

Carbon forms a large number of organic compounds because:

1. Catenation

Carbon can bond with itself to form long chains, branched chains and rings.

2. Tetravalency

Carbon has valency 4, so it can form four covalent bonds.

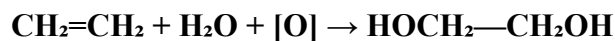
3. Strong covalent bonds

C—C and C—H bonds are strong due to small atomic size of carbon.

Therefore, carbon forms millions of stable organic compounds.

OR: Complete reactions

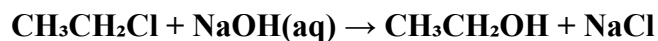
i. Ethene with cold dilute alkaline KMnO_4



Product: **ethylene glycol**

Reason: Cold dilute KMnO_4 adds two OH groups across double bond.

ii. Chloroethane with aqueous NaOH



Product: **ethanol**

Reason: OH^- replaces Cl^- .

iii. Ethene with hydrogen



Condition: Ni catalyst

Product: **ethane**

Q2(v) Azimuthal quantum number and its significance

Azimuthal quantum number is represented by **l**.

It determines:

1. Shape of orbital
2. Subshell of electron
3. Angular momentum of electron

Values of **l**:

For s-subshell: **l = 0**

For p-subshell: **l = 1**

For d-subshell: **l = 2**

For f-subshell: **l = 3**

Example: If $n = 3$, then l may be 0, 1, 2.

OR: Explain

i. Falling drop of water is spherical

Water drop is spherical due to **surface tension**.

Surface tension tries to reduce surface area. A sphere has minimum surface area for a given volume, so the drop becomes spherical.

ii. Water has higher surface tension than ether

Water molecules have strong hydrogen bonding.

Ether molecules have weaker intermolecular forces.

Therefore, water has higher surface tension than ether.

Q2(vi) Boiling points of CH_4 , CH_3CH_3 and H_2O

Given:

$\text{CH}_4 = -162^\circ\text{C}$

$\text{CH}_3\text{CH}_3 = -85^\circ\text{C}$

$\text{H}_2\text{O} = 100^\circ\text{C}$

Explanation

CH_4 and CH_3CH_3 are non-polar molecules. They have only London dispersion forces.

Ethane has greater molar mass than methane, so ethane has stronger dispersion forces and higher boiling point.

Water has hydrogen bonding, which is much stronger than dispersion forces.

Therefore:

$\text{CH}_4 < \text{CH}_3\text{CH}_3 < \text{H}_2\text{O}$

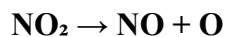
OR: Photochemical smog

Photochemical smog is brownish smog produced when sunlight reacts with nitrogen oxides and hydrocarbons in polluted air.

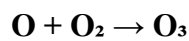
It mainly contains ozone, PAN and oxidants.

Formation

NO_2 absorbs sunlight:



Oxygen atom reacts with oxygen:

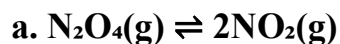


Ozone and hydrocarbons form photochemical smog.

It causes eye irritation, breathing problems and damage to plants.

Q2(vii) Entropy change

Entropy is the measure of randomness or disorder of a system.



One mole gas forms two moles gas.

Randomness increases.

So entropy **increases**.



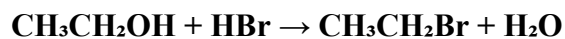
Two gaseous molecules form one solid.

Randomness decreases.

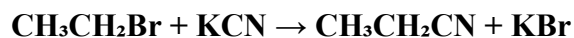
So entropy **decreases**.

OR: Ethanol to propanoic acid in three steps

Step 1: Ethanol to bromoethane



Step 2: Bromoethane to propanenitrile



Step 3: Propanenitrile to propanoic acid



Reason: KCN increases carbon chain by one carbon.

Q2 (viii) Methods to convert unusable water into clean water

1. **Filtration:** removes insoluble impurities like sand, clay and dust.
2. **Boiling:** kills harmful bacteria and germs.
3. **Chlorination:** chlorine is added to kill microorganisms and make water safe.

OR: Harmful effects of acid rain

1. It damages buildings and monuments, especially marble and limestone.
2. It makes lakes and rivers acidic, killing fish and aquatic life.
3. It damages plants by affecting leaves and soil nutrients.

Q2 (ix) Three methods for preparation of ethyl chloride from alcohol

Ethyl alcohol is $\text{C}_2\text{H}_5\text{OH}$ and ethyl chloride is $\text{C}_2\text{H}_5\text{Cl}$.

1. With HCl/ZnCl_2

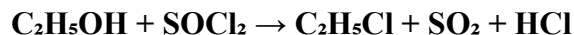


ZnCl_2 acts as a catalyst.

2. With PCl_5



3. With SOCl_2



This is a very good method because **SO_2 and HCl are gases**, so they escape and pure ethyl chloride is obtained.

OR: Lucas test for alcohols

Lucas reagent = **conc. HCl + anhydrous ZnCl_2**

It distinguishes alcohols by turbidity.

Alcohol	Observation
Tertiary alcohol	Immediate turbidity
Secondary alcohol	Turbidity after 5–10 minutes
Primary alcohol	No turbidity at room temperature

Reason: tertiary alcohol forms the most stable carbocation, so it reacts fastest.

Q2 (x) Stability of primary, secondary and tertiary carbocations

Carbocation has positive charge on carbon.

Stability order:

3° carbocation > 2° carbocation > 1° carbocation > methyl carbocation

Example:



Reason

Alkyl groups are electron-releasing groups. They push electron density towards the positively charged carbon by **+I effect** and **hyperconjugation**.

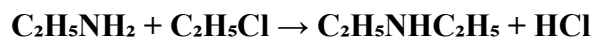
Tertiary carbocation has three alkyl groups, so it is most stable. Primary has only one alkyl group, so it is less stable.

OR: Reactions of ethyl amine

Ethyl amine:



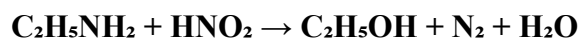
a. With ethyl chloride



Product: **diethylamine**

b. With HNO_2/HCl

Primary aliphatic amines react with nitrous acid to form alcohol and nitrogen gas.



Product: **ethanol**

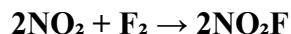
Q2 (xi) Metallic and non-metallic behavior

Metallic behavior	Non-metallic behavior
Metals lose electrons easily	Non-metals gain or share electrons
Low ionization energy	High ionization energy
Electropositive	Electronegative
Form basic oxides	Form acidic oxides
Example: Na, Mg, Ca	Example: O, Cl, S

Reason: Metals have a strong tendency to lose electrons, while non-metals have a strong tendency to gain electrons.

OR: Mechanism for reaction

Given reaction:



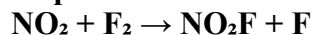
Rate law:

$$\text{Rate} = [\text{NO}_2][\text{F}_2]$$

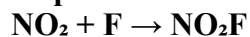
The slow step must contain **NO_2 and F_2** , because rate depends on both.

Proposed mechanism

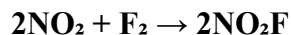
Step 1 slow:



Step 2 fast:



Adding both steps:



Reason: The rate-determining slow step gives:

$$\text{Rate} = k[\text{NO}_2][\text{F}_2]$$

which matches the given rate law.

Q2 (xii) Electronic configuration of Mn using $n + l$ rule and Hund's rule

Mn has atomic number:

$$Z = 25$$

Electronic configuration:

$$\text{Mn} = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$$

or

$$\text{Mn} = [\text{Ar}] 4s^2 3d^5$$

Application of $n + l$ rule

Orbitals with lower $n + l$ value fill first.

For 4s:

$$n = 4, l = 0$$

$$n + l = 4$$

For 3d:

$$n = 3, l = 2$$
$$n + l = 5$$

So **4s fills before 3d**.

Application of Hund's rule

In $3d^5$, five electrons occupy five d-orbitals singly:

3d: $\uparrow \uparrow \uparrow \uparrow \uparrow$

Reason: electrons first occupy equal-energy orbitals singly before pairing.

OR: Effect of catalyst on reaction rate

A catalyst increases the rate of reaction by providing an alternative pathway with lower activation energy.

It does not change:

ΔH , reactants, products, or equilibrium position.

Reaction pathway idea:

Without catalyst:

Reactants need **high activation energy**.

With catalyst:

Reactants need **low activation energy**.

So more molecules can cross the energy barrier, and reaction becomes faster.

Q2 (xiii) Concentration of H^+ in 1.0 M formic acid

Given:



$$K_a = 1.8 \times 10^{-4}$$

Initial concentration = **1.0 M**

Let:

$$[\text{H}^+] = x$$

Then:

$$[\text{HCOO}^-] = x$$

$$[\text{HCOOH}] \approx 1.0$$

Ka expression:

$$K_a = [\text{H}^+][\text{HCOO}^-] / [\text{HCOOH}]$$

So:

$$1.8 \times 10^{-4} = x^2 / 1.0$$

$$x^2 = 1.8 \times 10^{-4}$$

$$x = \sqrt{(1.8 \times 10^{-4})}$$

$$x = 1.34 \times 10^{-2} \text{ M}$$

So:

$$[\text{H}^+] = 1.34 \times 10^{-2} \text{ M}$$

OR: Iodoform test for aldehydes and ketones

Iodoform test is used to identify compounds having:

CH₃CO— group

or alcohols oxidizable to this group.

Positive test gives yellow precipitate of **CHI₃**.

For aldehyde

Acetaldehyde gives positive test:



Yellow ppt = **iodoform**

Formaldehyde does not give this test.

For ketone

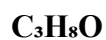
Methyl ketones give positive test:



So acetone gives yellow precipitate of iodoform.

Q2 (xiv) Position isomers and functional group isomers of $\text{C}_3\text{H}_7\text{OH}$

Molecular formula:



Position isomers of alcohol

These have same functional group but OH position is different.

1. Propan-1-ol



2. Propan-2-ol



Functional group isomer

Alcohol and ether can be functional group isomers.

Alcohol:



Ether:



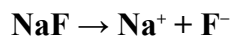
Name: **methoxyethane**

OR: Why ionization of HF decreases when its salt is added?

HF is a weak acid:



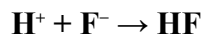
When its salt, for example **NaF**, is added:



The concentration of F^- increases.

According to Le Chatelier's principle, when product concentration increases, equilibrium shifts backward to reduce the change.

So:



Therefore, ionization of HF decreases.

This is called the **common ion effect**.

Section C — Long Questions

Q3

Hybridization

Hybridization is the mixing of atomic orbitals of nearly equal energy to form new equivalent orbitals called hybrid orbitals.

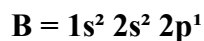
Three differences between sp^3 and sp^2 hybridization

sp^3 hybridization	sp^2 hybridization
One s and three p orbitals mix	One s and two p orbitals mix
Forms 4 hybrid orbitals	Forms 3 hybrid orbitals
Shape is tetrahedral	Shape is trigonal planar
Bond angle is 109.5°	Bond angle is 120°
Example: CH_4	Example: BF_3

Bonding in BF₃ on the basis of hybridization

Boron has atomic number 5.

Electronic configuration:



In BF₃, boron forms three bonds with three fluorine atoms. One 2s and two 2p orbitals of boron mix to form **three sp² hybrid orbitals**.

These three sp² orbitals overlap with orbitals of fluorine atoms and form three **B—F sigma bonds**.

BF₃ has:

Shape: trigonal planar

Bond angle: 120°

Reason: three bond pairs arrange themselves as far apart as possible in one plane.

OR: Why are alkanes chemically less reactive?

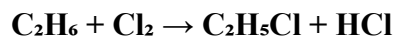
Alkanes are less reactive because:

1. They contain only **C—C and C—H sigma bonds**. Sigma bonds are strong.
2. They are **non-polar**, so electrophiles and nucleophiles do not attack them easily.
3. Carbon and hydrogen have almost similar electronegativity, so C—H bond is nearly non-polar.
4. They are saturated hydrocarbons, so they do not normally undergo addition reactions.

That is why alkanes mostly react only under strong conditions, such as sunlight or high temperature.

Free radical chlorination of ethane to form ethyl chloride

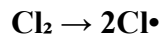
Overall reaction:



Condition: **sunlight / UV light**

Step 1: Initiation

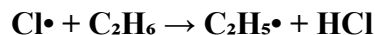
Chlorine molecule breaks into chlorine free radicals.



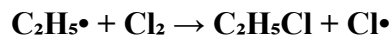
Reason: UV light breaks the Cl—Cl bond homolytically.

Step 2: Propagation

Chlorine radical attacks ethane:



Ethyl radical then reacts with chlorine:



The chlorine radical is regenerated, so chain reaction continues.

Step 3: Termination

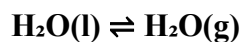
Free radicals combine and reaction stops.



Q4

Physical method to recognize equilibrium

A common physical equilibrium is evaporation and condensation of water in a closed container:



At equilibrium:

Rate of evaporation = rate of condensation.

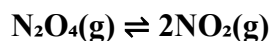
Observation:

The level of liquid water remains constant in a closed container.

Reason: water molecules are still evaporating and condensing, but both rates are equal.

Chemical method to recognize equilibrium

Example:



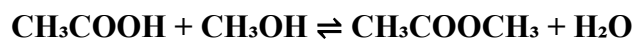
N_2O_4 is colorless, while NO_2 is brown.

At equilibrium, the brown color becomes constant.

Reason: forward and reverse reactions continue at equal rates, so concentrations of reactants and products remain constant.

OR: Reaction between CH_3COOH and CH_3OH

This is esterification.



Product:

Methyl ethanoate / methyl acetate

Conditions:

Concentrated H_2SO_4 and heat

Concentrated H_2SO_4 acts as an acid catalyst and dehydrating agent.

Mechanism of esterification

Step 1: Protonation of carboxylic acid

Oxygen of carbonyl group accepts H^+ from acid.

This makes carbonyl carbon more positive and more reactive.

Step 2: Attack by methanol

Methanol attacks the positive carbonyl carbon.

Step 3: Proton transfer

Hydrogen shifts inside the molecule to make water a good leaving group.

Step 4: Removal of water

Water molecule leaves.

Step 5: Deprotonation

H⁺ is removed and ester is formed.

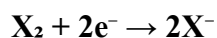
Final product:



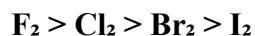
Q5

Relative reactivity of halogens as oxidizing agents

Halogens act as oxidizing agents because they gain electrons.



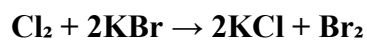
Reactivity order:



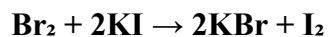
Reason: fluorine gains electrons most easily, so it is the strongest oxidizing agent.

Examples

Chlorine displaces bromine from bromide solution:



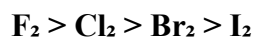
Bromine displaces iodine from iodide solution:



But iodine cannot displace bromine or chlorine because it is weaker oxidizing agent.

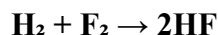
Relative reactivity of halogens with H₂

Order:

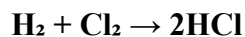


Reactions

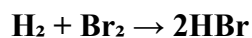
Fluorine reacts explosively even in dark:



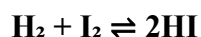
Chlorine reacts in sunlight:



Bromine reacts slowly on heating:



Iodine reacts very slowly and reversibly:



Reason: oxidizing power decreases down the group and bond strength/ease of reaction decreases.

OR: Behaviour of 1-propanol and 2-propanol

(i) Reaction with I₂/NaOH

This is iodoform test.

2-propanol gives positive iodoform test because it has the group:



Reaction:



Observation:

Yellow precipitate of **CHI₃**.

1-propanol does not give iodoform test.

Reason: it does not contain $\text{CH}_3\text{CH}(\text{OH})\text{—}$ group.

(ii) Reaction with HCl/ZnCl_2

This is Lucas test.

2-propanol is a secondary alcohol, so it gives turbidity after a few minutes.



Product:

2-chloropropane

1-propanol is a primary alcohol, so it gives no turbidity at room temperature.



Product:

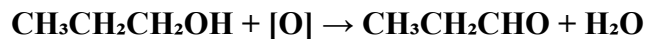
1-chloropropane

Reason: secondary alcohol forms more stable carbocation than primary alcohol.

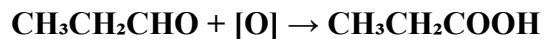
(iii) Reaction with $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$

This is oxidation.

1-propanol is primary alcohol, so it oxidizes first to aldehyde and then to acid.



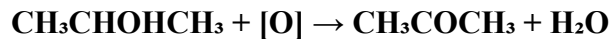
Further oxidation:



Final product:

propanoic acid

2-propanol is secondary alcohol, so it oxidizes to ketone.



Product:

propanone / acetone

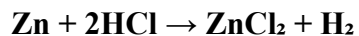
Q6

Given:

Mass of Zn = 300 g

Mass of HCl = 300 g

Reaction:



Molar mass:

Zn = 65.38 g/mol

HCl = 1 + 35.5 = 36.5 g/mol

H₂ = 2 g/mol

Step 1: Calculate moles

Moles of Zn:

$$300 / 65.38 = 4.59 \text{ mol}$$

Moles of HCl:

$$300 / 36.5 = 8.22 \text{ mol}$$

Step 2: Find limiting reactant

Reaction says:

1 mol Zn needs 2 mol HCl.

For 4.59 mol Zn, HCl needed:

$$4.59 \times 2 = 9.18 \text{ mol HCl}$$

But available HCl = **8.22 mol**

So **HCl is limiting reactant.**

Step 3: Calculate theoretical moles of H₂

From equation:

2 mol HCl produce 1 mol H₂.

So:

8.22 mol HCl produce $8.22 / 2 = 4.11$ mol H₂

Mass of H₂:

$$4.11 \times 2 = 8.22 \text{ g}$$

Theoretical yield of H₂ = **8.22 g**

Step 4: Percentage yield

Actual yield = 6 g

Percentage yield = Actual yield / Theoretical yield $\times$ 100

$$= 6 / 8.22 \times 100$$

$$= 72.99\%$$

Final answer:

Mass of H₂ produced theoretically = 8.22 g

Percentage yield = 73% approximately

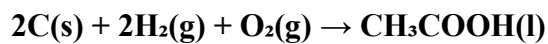
OR: Hess's law and enthalpy of formation of CH₃COOH

Hess's law

Hess's law states that the total enthalpy change of a reaction is the same whether the reaction takes place in one step or several steps, provided initial and final states are same.

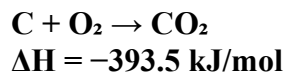
Required reaction

Formation of acetic acid:

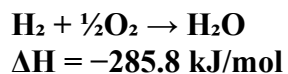


We are given combustion enthalpies:

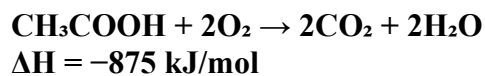
Carbon:



Hydrogen:

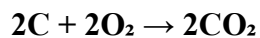


Acetic acid:



Step 1: Combustion of elements for formation equation

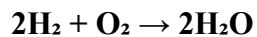
For 2C:



ΔH :

$$2 \times -393.5 = -787.0 \text{ kJ}$$

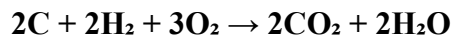
For 2H₂:



ΔH :

$$2 \times -285.8 = -571.6 \text{ kJ}$$

Total:

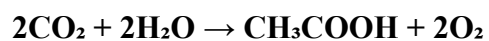


ΔH :

$$-787.0 - 571.6 = -1358.6 \text{ kJ}$$

Step 2: Reverse combustion of acetic acid

Reverse:

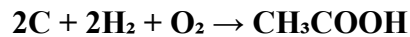


ΔH :

$$+875 \text{ kJ}$$

Step 3: Add equations

After cancellation:



Total ΔH :

$$-1358.6 + 875$$

$$= -483.6 \text{ kJ/mol}$$

Final answer:

$$\text{Enthalpy of formation of CH}_3\text{COOH} = -483.6 \text{ kJ/mol}$$